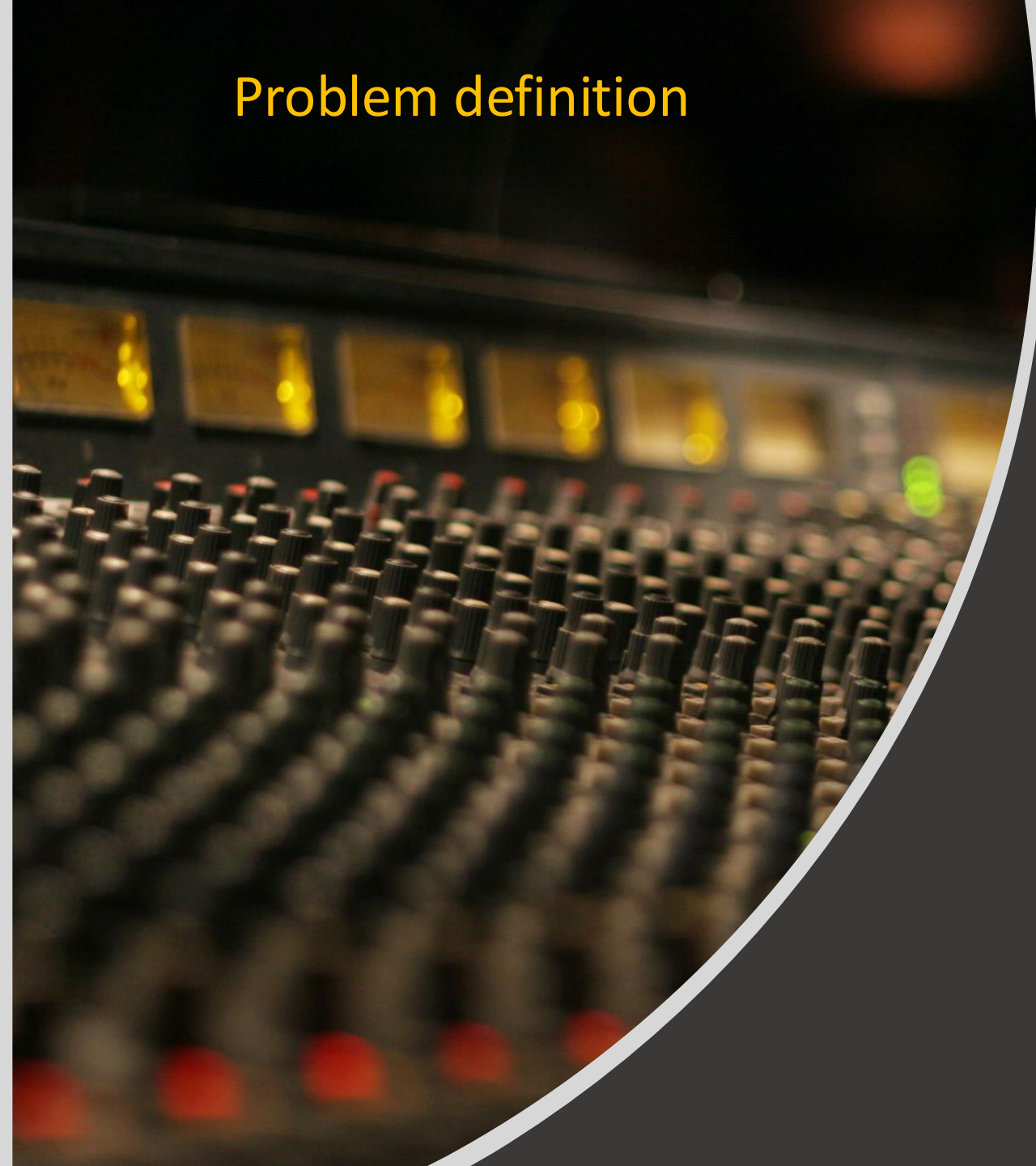




Music Recommendation System

Marta Wheeler

Problem definition



Recommendations Systems are one of the most profitable applications of data science in today's world.

Market size value in **2022** -> **\$34.53** billion
Revenue forecast in **2030** -> **\$103.07** billion
Growth rate CAGR of **14.7%** from 2022 to 2030

Spotify, YouTube Music, Pandora, SoundCloud, iHeartRadio, Tidal, Tidal, Apple Music, Bandcamp
Spotify's market cap is estimated at **\$24.08 billion**



Problem to solve

Increased customer satisfaction

Increased revenue

Find out how to measure customer happiness
Engagements & retention

Recommend top 10 songs for a user of music
recommendation system

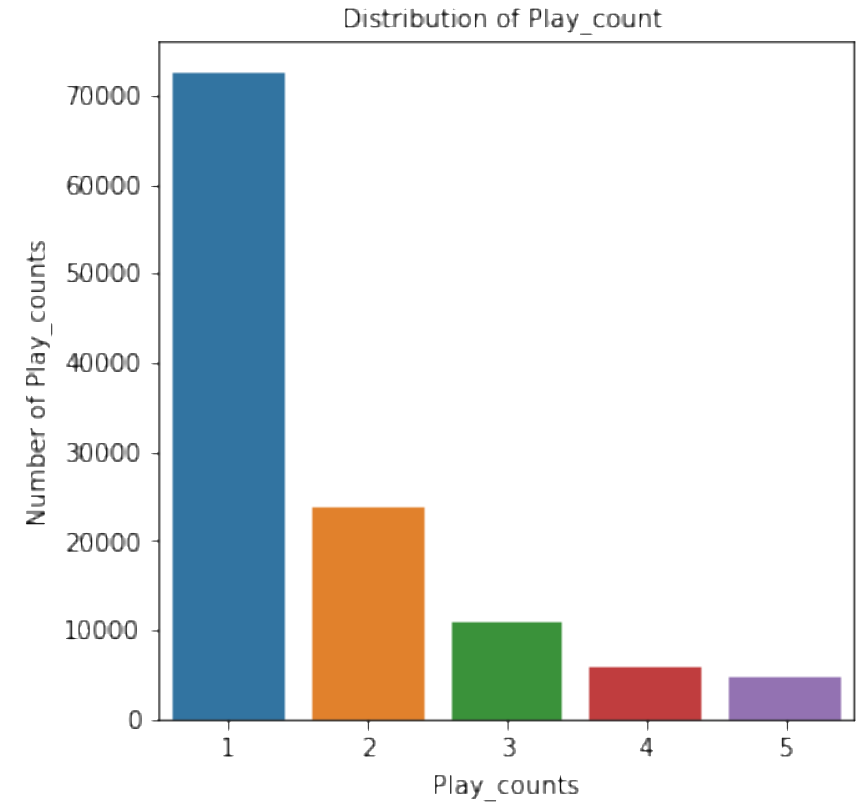


Unique characteristics of music recommendation:

- Duration of items – short
- Magnitude of items – large
- Sequential consumption
- Recommendation of previously recommended items
- Consumption behavior – passive
- Listening intent and purpose
- Emotions
- Listening context

Solution Approach

- Choosing only **subset** of the data - Taste Profile Subset released by the Echo Nest as part of the Million Song Dataset
 - only users who listened to at least **90** or more songs
 - songs that have been listened by **120** or more users
 - only songs with max play count **5**
- **Play count** as a **proxy** for rating
- Threshold = 1.5
- Building several different models and comparing its evaluation metrics

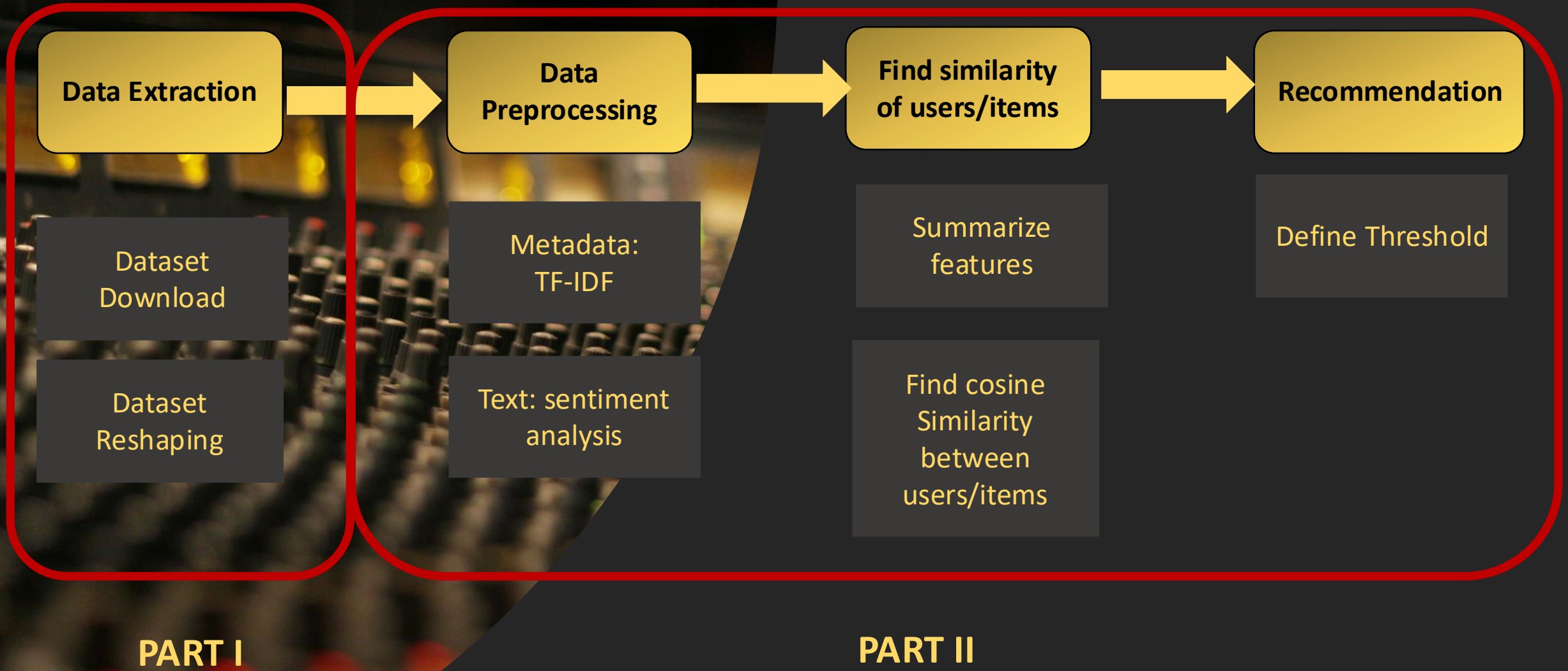




Investigated models:

- Popularity based
- User-User Similarity-based collaborative filtering
- Item-item Similarity-based collaborative filtering
- Matrix factorization – model based
- Cluster based
- Content based

Solution Approach - Recommendation System Pipeline



More details on solution approach

Perf eval measures	Similarity Based Collaborative Filtering				Model Based Collaborative Filtering		Cluster based Recommendation System	
	User - user		Item - item		Matrix Factorization			
	Baseline	Optimized	Baseline	Optimized	Baseline	Optimized	Baseline	Optimized
RMSE	1.0878	1.0521	1.0394	1.0328	1.0252	1.0141	1.0691	1.0487
Precision	0.396	0.413	0.307	0.408	0.41	0.415	0.398	0.397
Recall	0.692	0.721	0.562	0.665	0.633	0.635	0.56	0.582
F1	0.504	0.525	0.397	0.506	0.498	0.502	0.465	0.472

User_id	Song_id								
6958	1671	Pred1	1.801	1.963	1.361	1.963	1.267	1.343	1.294 1.294
6958	3232	Pred2	1.639	1.452	1.378	1.492	1.556	1.804	1.479 1.479
27018	1671	Pred3	1.275	1.286	2.551	2.337	2.217	2.264	2.403 2.403

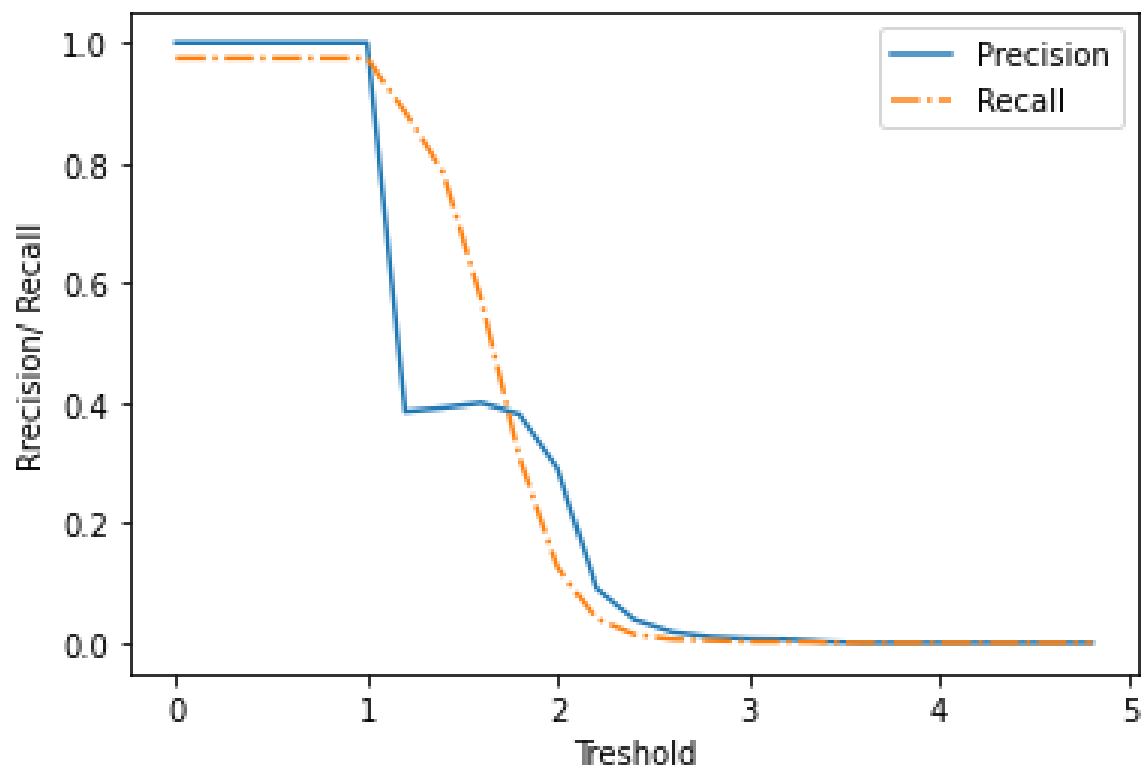


Final Model solution existing users and songs

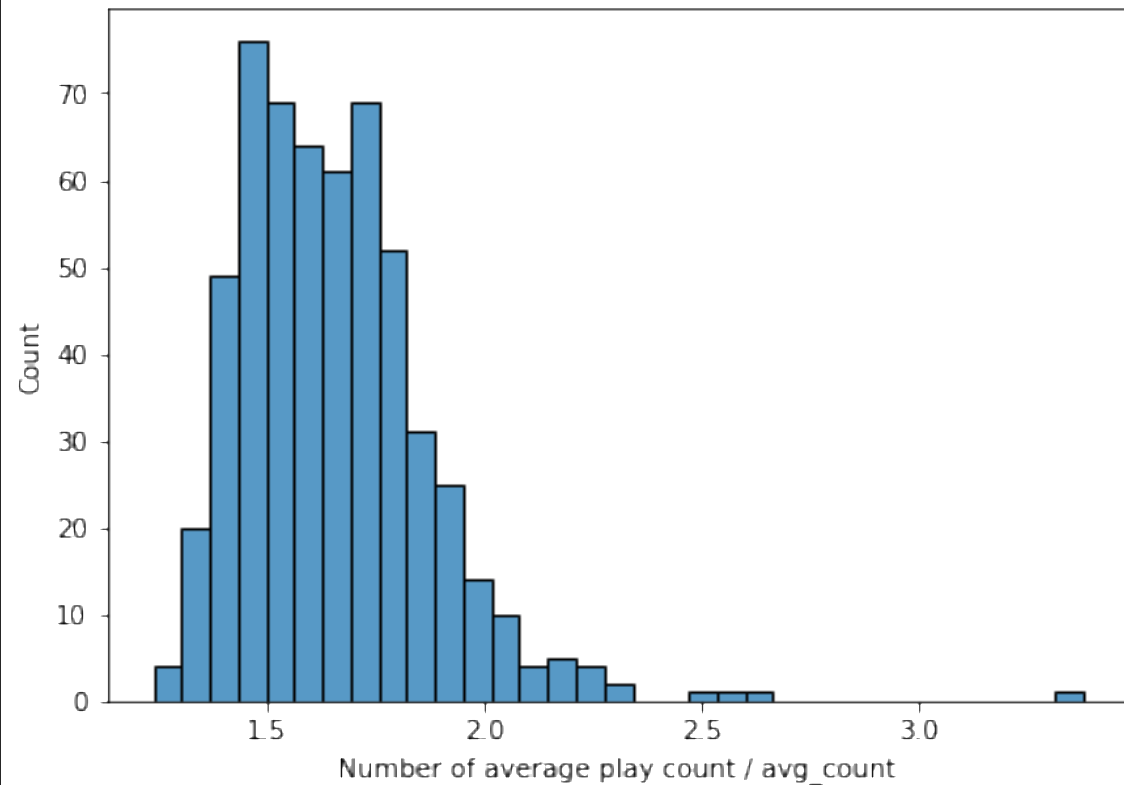
- **Optimized user-user similarity based collaborative filtering model.**
- **Optimized by tuning values of the hyperparameters using grid search cross-validation**
 - **k = 30**
 - **min_k = 9**
 - **sim_options**
 - name = 'pearson_baseline'
 - user_based = True
 - min_support 2
- **Test size = 40%**
- **Threshold = 1.5**

Model solution – threshold

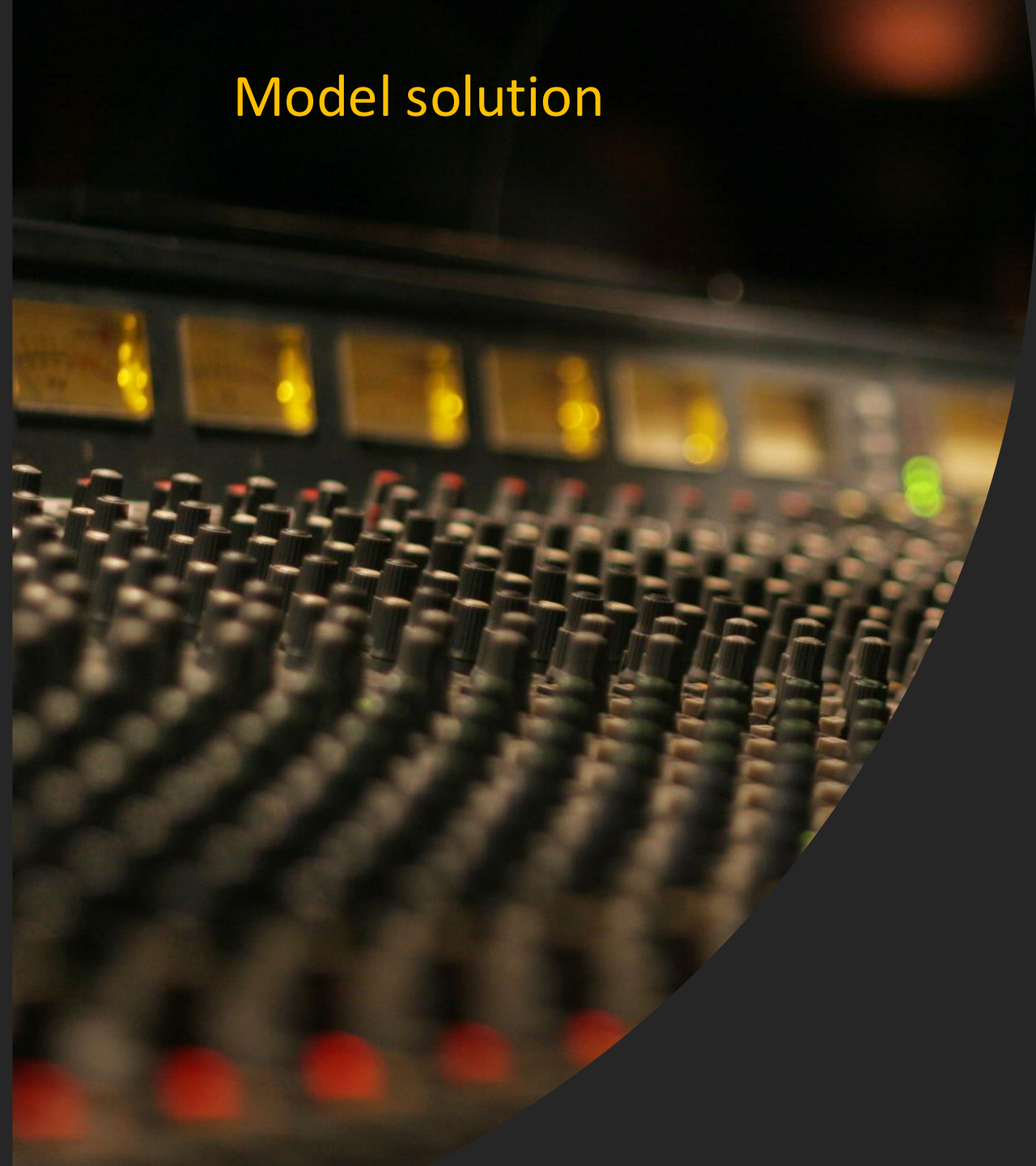
Precision & Recall for different Treshold



Histogram of avg_count



Model solution



Existing users and songs vs new ones.

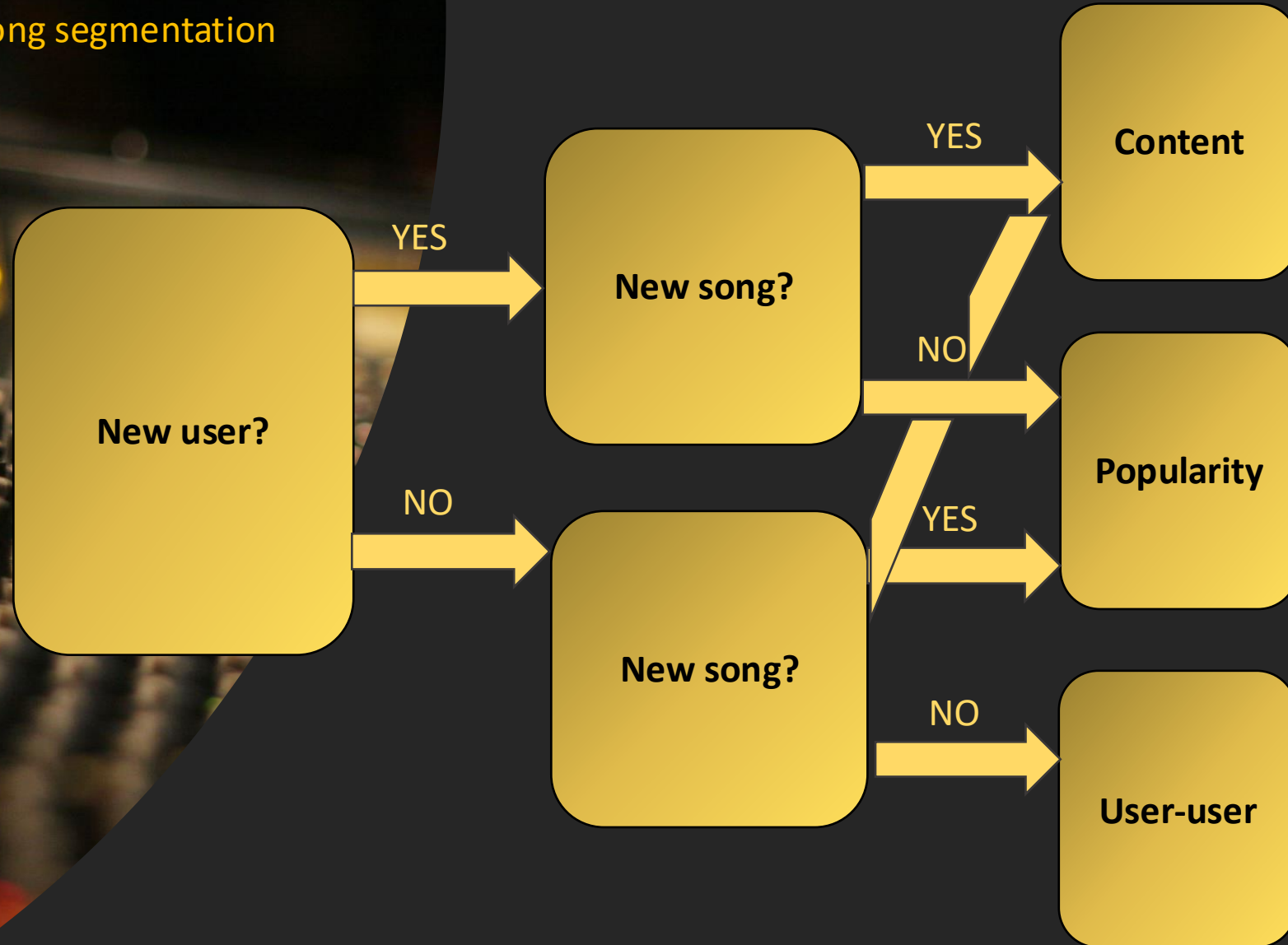
Optimized User-user similarity based collaborative filtering for established users and songs.

Popularity based model to introduce **new** songs to existing users (**item** cold start problem).

Content based models for brand new songs and artist (**user** cold start problem).

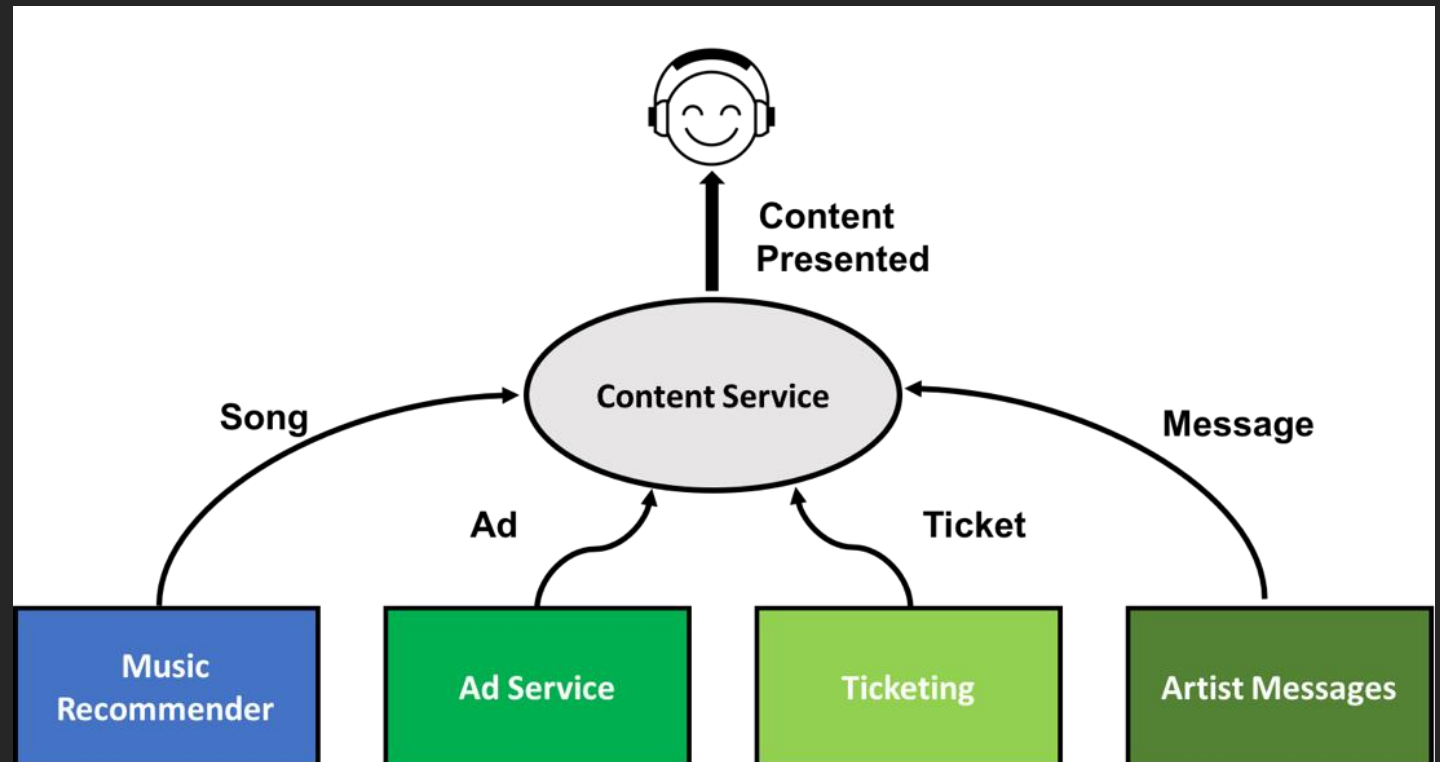
Proposed business solution

Hybrid model user and song segmentation



Key Actionable for stakeholders

- Customer satisfaction
- Engagement vs Retention
- How to measure user happiness?
- How to generate max profit?
- How to reduce cost?





Executive Summary

- Using **hybrid model** to recommend 10 songs for a user
 - **User-user** similarity-based model for **existing users**
 - **Content** based model for new songs and **new users**
 - **Popularity** based model to suggest **new songs** for **existing users**
- **User-user model** exhibits best evaluation measures (high Precision@k and Recall@k and low RMSE)
- **Low precision** yields in **high cost** (60% of recommended songs aren't relevant)
- Having richer data set can help with better prediction and lower cost
- Measuring customer satisfaction is an important aspect



Risk and Challenges

- Finding the right song for the right user
 - New song vs repeated
 - User satisfaction evaluation
- Cold start & sparsity problem
 - Content based recommendation
 - Hybridization (CB + CF)
 - Cross-domain recommendation
 - Active learning
- Automatic playlist continuation
- High cost of recommendations compared to return



Cold Start Challenge & User Happiness

- Cold Start Problem
- Sparsity Problem

More data need for better models

- **Users** – typical demographical data (age, geo location, education level, income, ethnicity), new user vs established user, free or paid membership.
- **Songs** – genre, type, sonic info (acoustic features including spectral properties, timbre, rhythm, and pitch).
- **Interactions** – seasonality (time of the year, time of the weekday), place (gym, plane, home- living room, entertainment, kitchen, bedroom, car, work), activity- can be tight to the place (resting/relaxing, working out, work, travel, cooking), time spent listening to the song (if the song was listened to till the end, stopped at 10%, 30%, skipped and how many times).



Further analysis

- Further analysis would consist of acquiring **additional data** mentioned previously (user, song and interaction data) and finding any **dependencies** (for example seasonality, time and place for an interaction, more context analysis for the songs and users).
- Having richer data, different models could be applied like **time series (seasonality)** or **neural networks (CRNN – combination of CNN & RNN)**



Thank you!

Bibliography

- [1] Schedl Markus, Zamani Hamed, Chen Ching-Wei, Daldjoo Yashar, Elahi Medhi (2018) Current challenges and vision in music recommender system research.
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- [3] Adiyansjah, Gunawan Alexander, Suhartono Derwin (2019) Music Recomemndaer System Based on Genre unising Convolutional Recurrent Neural Networks
- [4] Grand View Research (2021) Music Streaming Market.



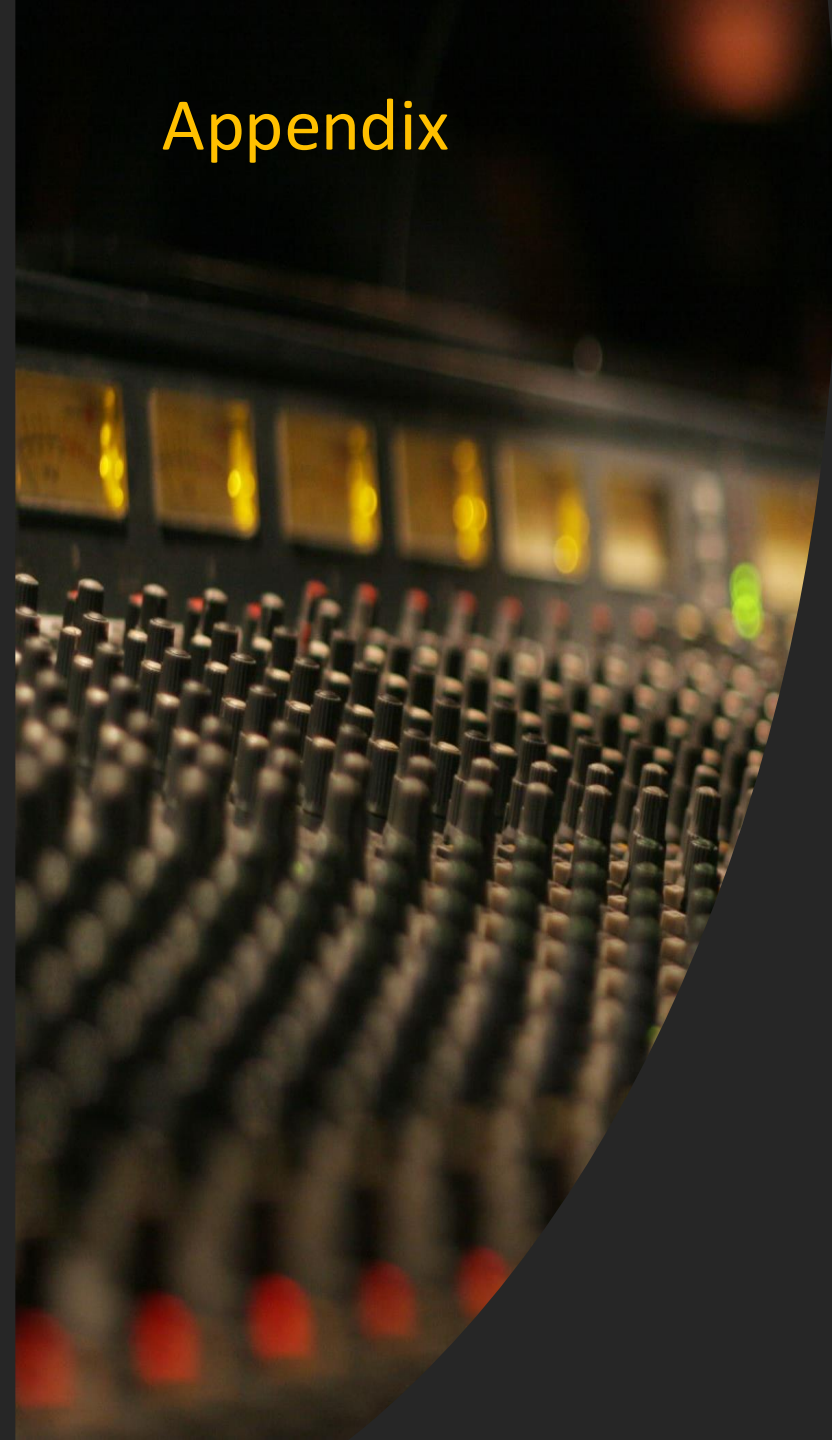
Appendix



Appendix

Hyperparameters for the KNNBasic algorithms

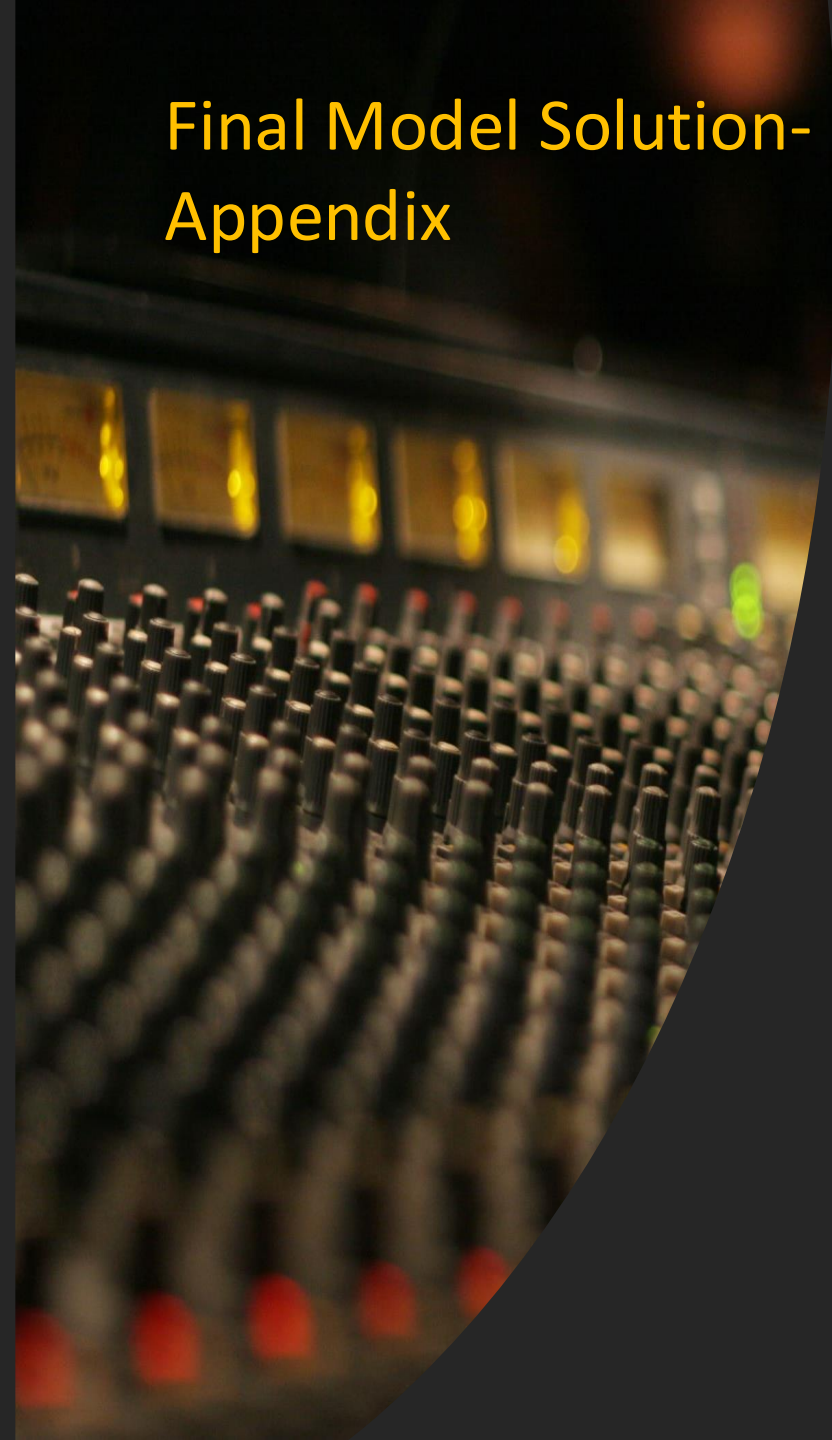
- **k** (int) – The (max) number of neighbors to take into account for aggregation (see this note). Default is 40.
- **min_k** (int) – The minimum number of neighbors to take into account for aggregation. If there are not enough neighbors, the prediction is set to the global mean of all ratings. Default is 1.
- **sim_options** (dict) – A dictionary of options for the similarity measure. And there are four similarity measures available in surprise -
 - cosine
 - msd (default)
 - pearson
 - pearson baseline



Appendix

**10 most recommended songs to the user number 6958
(user-user similarity based collaborative filtering model):**

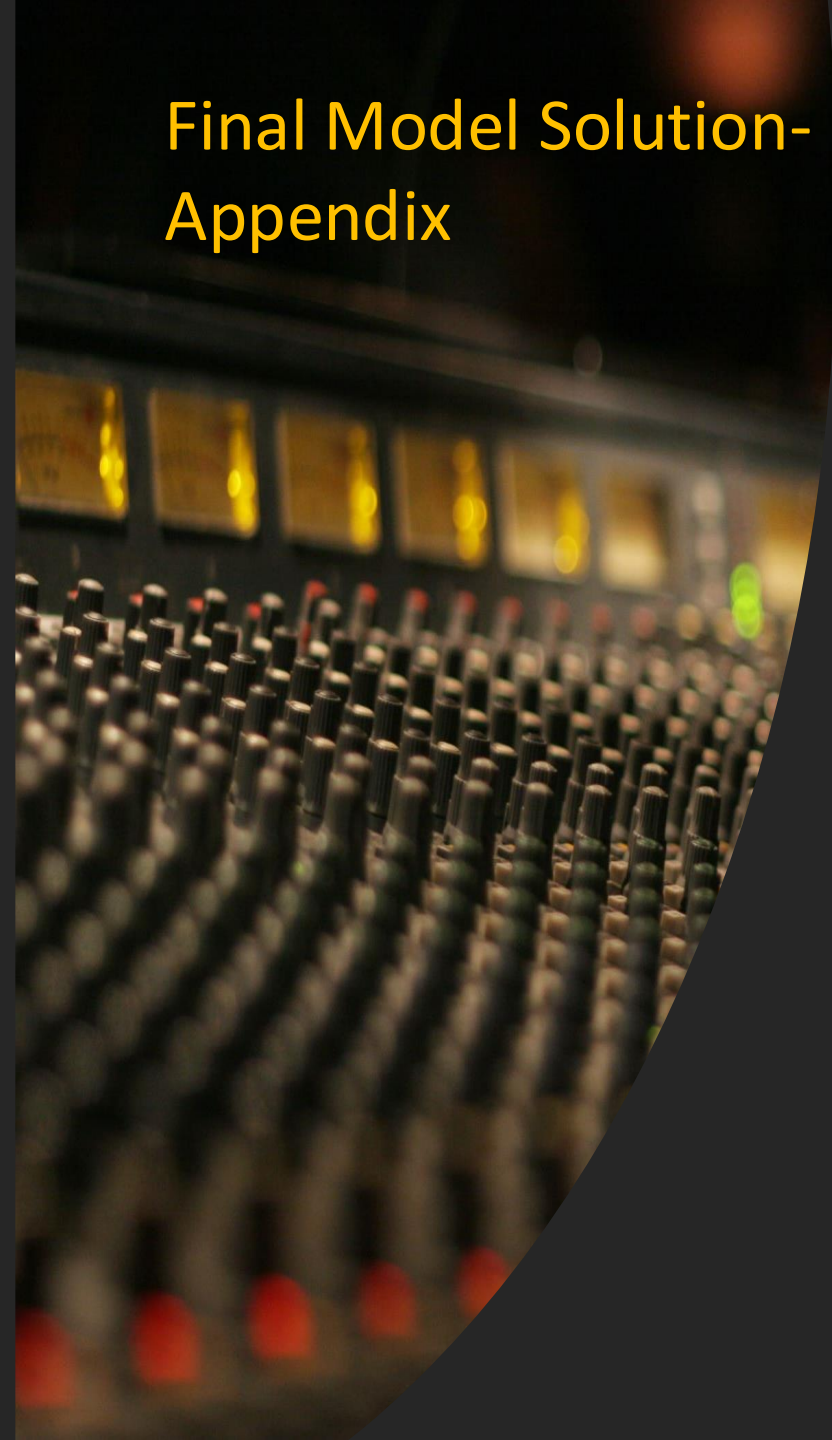
1. 2.55 5531 "Secrets" Waking Up by OneRepublic from 2009
2. 2.52 317 "Undo" Vespertine Live by Björk from 2001
3. 2.41 4954 "The Maestro" Check Your Head by Beastie Boys from 1992
4. 2.4 8635 "Una Confusion" All Access by LU from 2006
5. 2.39 5943 "You've Got The Love" Lungs by Florence + The Machine from 2009
6. 2.39 1664 "Horn Concerto No. 4 in E flat K495: II. Romanc..."
Mozart - Eine kleine Nachtmusik by Barry Tuckwell/Academy of St Martin-in-the-Fie...
7. 2.35 6246 "Canada" The End Is Here by Five Iron Frenzy
8. 2.32 1348 "Mia" First Love by Emmy The Great from 2007
9. 2.27 7496 "The Gift" We Don't Need To Whisper by Angels and Airwaves from 2006
10. 2.26 2852 "West One (Shine On Me)" The Crack/Grin And Bear It by The Ruts from 1980



Final Model Solution- Appendix

Popularity based model with min 50 interactions:

1. 7224 "Victoria (LP Version)" Hit By A Train: The Best Of Old 97's by Old 97's from 2006
 2. 8324 "The Big Gundown" Invaders Must Die Remixes and Bonus Tracks by The Prodigy from 2009
 3. 6450 "Brave The Elements" Brave The Elements - EP by Colossal
 4. 9942 "Greece 2000" Greece 2000 by Three Drives from 1997
 5. 5531 "Secrets" Waking Up by OneRepublic from 2009
 6. 5653 "Transparency" Workout Holiday by White Denim from 2008
 7. 8483 "Video Killed The Radio Star" Friends Reunited: Music Of The Year 1979 by The Buggles from 1979
 8. 2220 "Sehr kosmisch" Musik von Harmonia by Harmonia
 9. 657 "Luvstruck" Hard House Anthems by Southside Spinners from 1999
- 614 "You're The One" If There Was A Way by Dwight Yoakam from 1990



Final Model Solution- Appendix

**10 most recommended songs to the user number 6958
(content-based model, because user listened to the “The Bachelor and the Bride” 5 times)**

- 1. 447 “Daisy And Prudence” Distillation by Erin McKeown from 2000**
- 2. 1674 “I Need A Dollar” I Need A Dollar by Aloe Blacc from 2010**
- 3. 1936 “Feel The Love” In Ghost Colours by Cut Copy from 2008**
- 4. 2716 “All The Pretty Faces” When You Were Young by The Killers from 2006**
- 5. 3534 “Who Let You Go?” Sawdust by The Killers from 2007**
- 6. 5095 “Bones” Bones by The Killers from 2006**
- 7. 5715 “Sam's Town” Sam's Town by The Killers from 2006**
- 8. 8624 “Hearts On Fire” Hearts On Fire by Cut Copy from 2007**
- 9. 2914 “Billy Liar” Billy Liar (CD-Single) by The Decemberists from 2003**
- 10. 6084 “Red Right Ankle” Her Majesty The Decemberists by The Decemberists from 2003**